MLOPS LifeCycle



Machine learning



- Build
- Train
- Deploy
- Monitor
- Manage
- Re-train

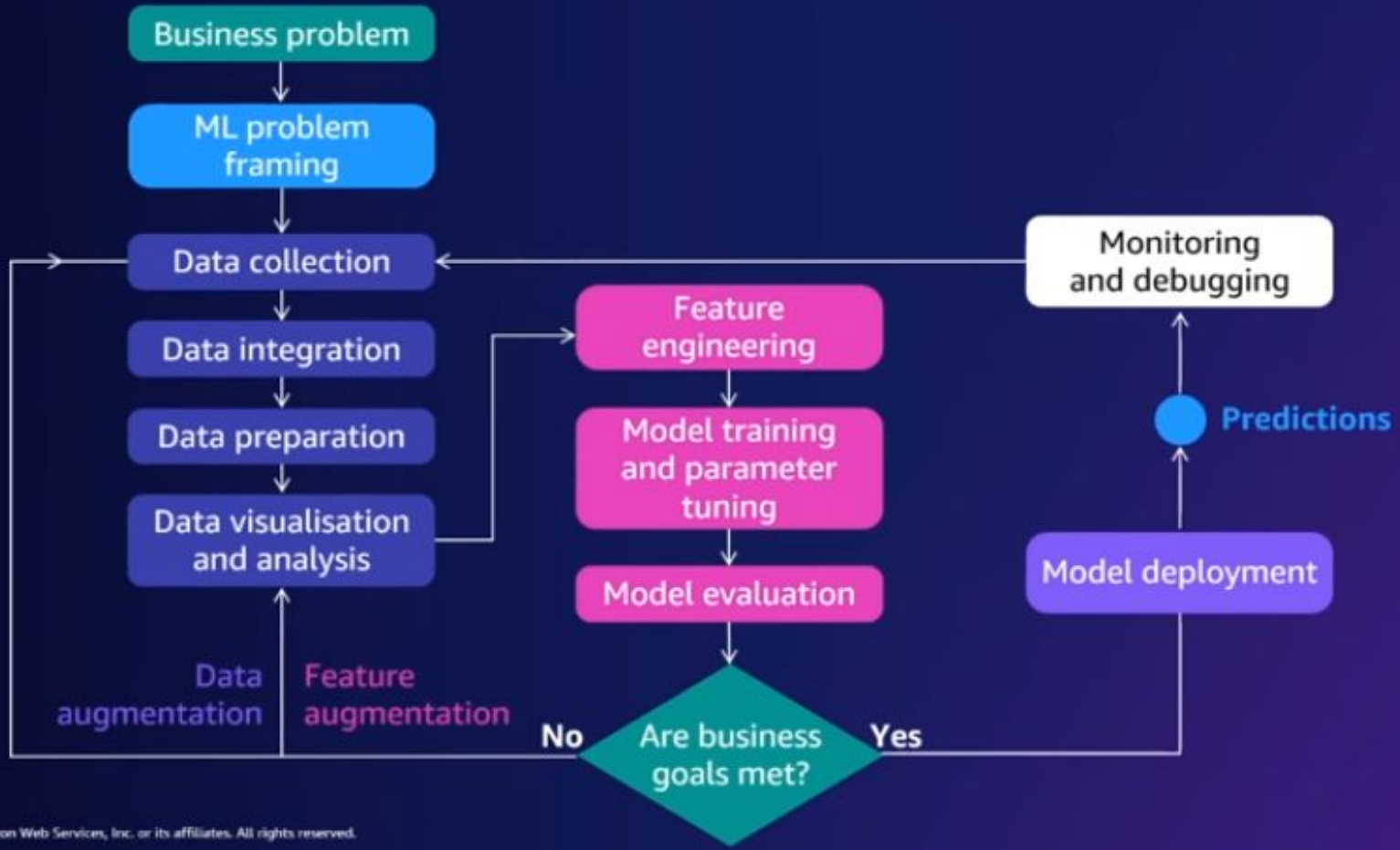


Manual workflows can become bottlenecks, impacting productivity, performance, and costs

Innovation



Machine learning is an iterative process



aws
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1. Problem Framing

- Goal: Define the business problem you're trying to solve using machine learning and establish the desired outcome.
- Example (Retail): Predicting customer churn to develop targeted retention campaigns.
- Example (Healthcare): Identifying patients at high risk of readmission to optimize care plans.

2. Data Collection

- Goal: Gather relevant data from various sources to train the machine learning model.
- Example (Retail): Sales transaction data, customer demographics, website clickstream data.
- Example (Finance): Stock market data, economic indicators, loan application data.

3. Data Integration

- Goal: Combine data from different sources into a unified format for analysis.
- Example (Retail): Merging sales data with customer loyalty program information.
- Example (Finance): Integrating stock price data with economic news sentiment.
- Tools and Cloud services:AWS Glue, Azure Data Factory, GCP Dataflow ETL, ELT

4. Data Preparation

- Goal: Clean and pre-process the data to ensure its quality and suitability for modelling.
- Example (Retail): Handling missing values in customer data, identifying and removing outliers in sales figures.
- Example (Healthcare): Standardizing medical records format, addressing inconsistencies in patient data.

5. Data Visualization and Analysis

- Goal: Explore and understand the data to identify patterns and trends that can inform feature engineering and model selection.
- Example (Retail): Visualizing customer purchase behaviour over time, analyzing correlations between product categories and demographics.
- Example (Finance): Creating charts to visualize historical stock price movements, identifying relationships between economic indicators and loan defaults.

6. Feature Engineering

- Goal: Create new features from existing data that improve the model's learning ability.
- Example (Retail): Deriving features like average purchase value per customer or product category groupings.
- Example (Healthcare): Extracting features from medical images or creating new variables based on patient medical history.

7. Model Training and Tuning

Goal: Train a machine learning model on the prepared data and fine-tune its hyperparameters for optimal performance.
Example (Retail): Training a classification model to predict customer churn based on purchase history.
Example (Finance): Training a regression model to estimate stock prices or a classification model to assess loan default risks.

8. Model Evaluation

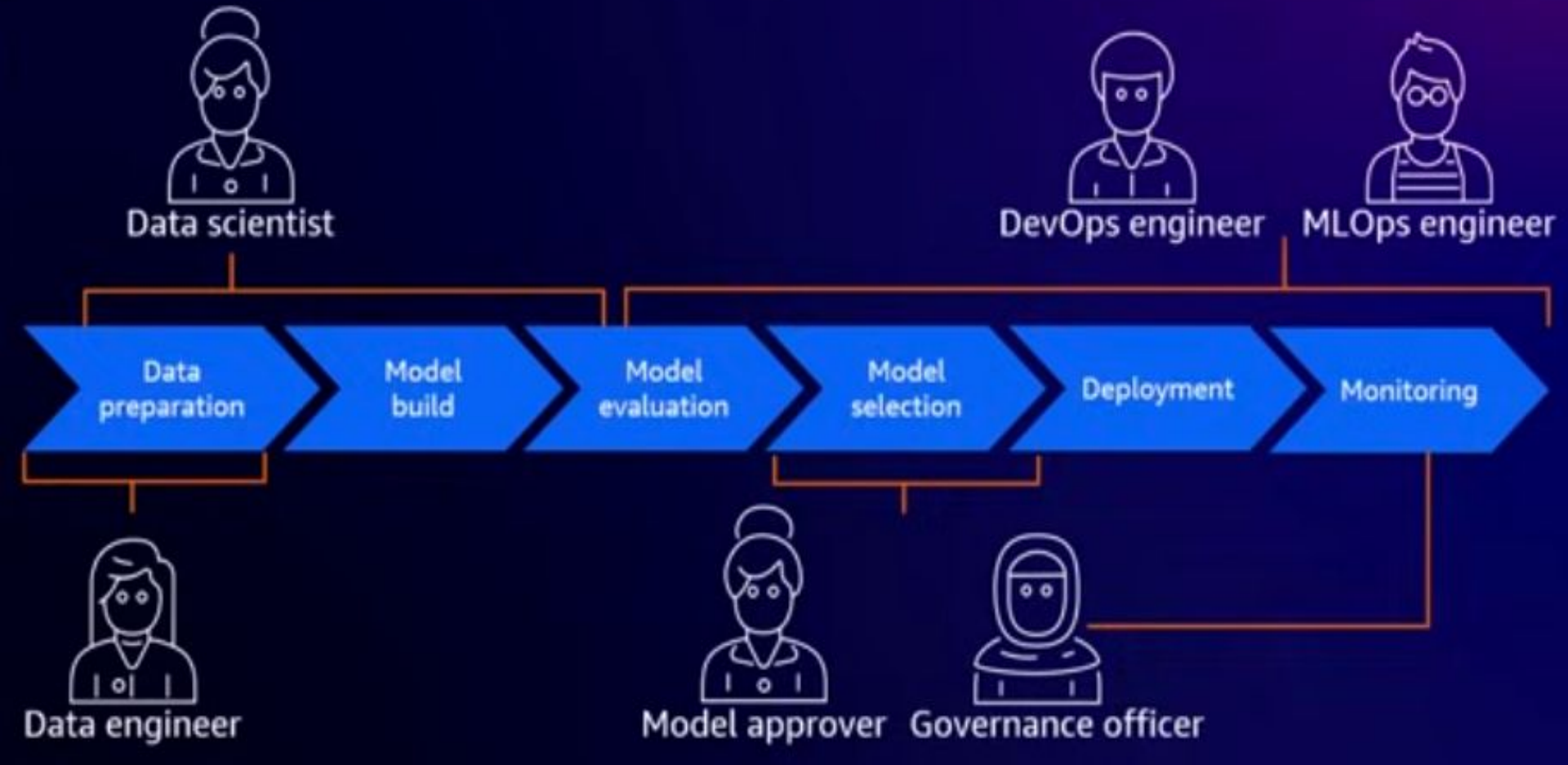
Goal: Assess the performance of the trained model on unseen data to measure its effectiveness and identify potential shortcomings.
Example (Retail): Evaluating the accuracy of the churn prediction model using metrics like confusion matrix or AUC-ROC curve.
Example (Healthcare): Measuring the precision and recall of the disease diagnosis model to assess its ability to correctly classify patients.

9. Feature Augmentation (Optional)

Goal: Artificially create new variations of existing features to improve model robustness and generalizability.
Example (Retail): Generating synthetic customer profiles to address data imbalance or scarcity.
Example (Healthcare): Creating slightly modified medical images to improve the generalizability of a disease classification model.

10. Data Augmentation (Optional)
Goal: Artificially create new data samples to increase the size and diversity of the training data.
Example (Retail): Simulating additional customer purchase scenarios to enrich the training data.
Example (Finance): Generating synthetic stock market data to improve the model's ability to handle unseen market conditions.

ML workflow and associated roles



	SERVICE TYPE	DESCRIPTION	aws	Azure	Google Cloud
STORAGE	Object storage	For storing any files you regularly use	Simple Storage Service (S3)	Blob Storage	Cloud Storage Buckets
	Archive storage	Low cost (but slower) storage for rarely used files	S3 Glacier Instant, Glacier Flexible, Glacier	Blob Cool/Cold/Archive tiers	Cloud Storage Nearline, Coldline, Archive
	File storage	For storing files needing hierarchical organization	Elastic File System (EFS), FSx	Avers vFXT, Files	Filestore
	Block storage	For storing groups of related files	Elastic Block Storage	Disk Storage	Persistent Disk
	Hybrid storage	Move files between on-prem & cloud	Storage Gateway	StorSimple, Migrate	Storage Transfer Service
	Edge/offline storage	Move offline data to the cloud	Snowball	Data Box	Transfer Appliance
	Backup	Prevent data loss	Backup	Backup	Backup and Disaster Recovery

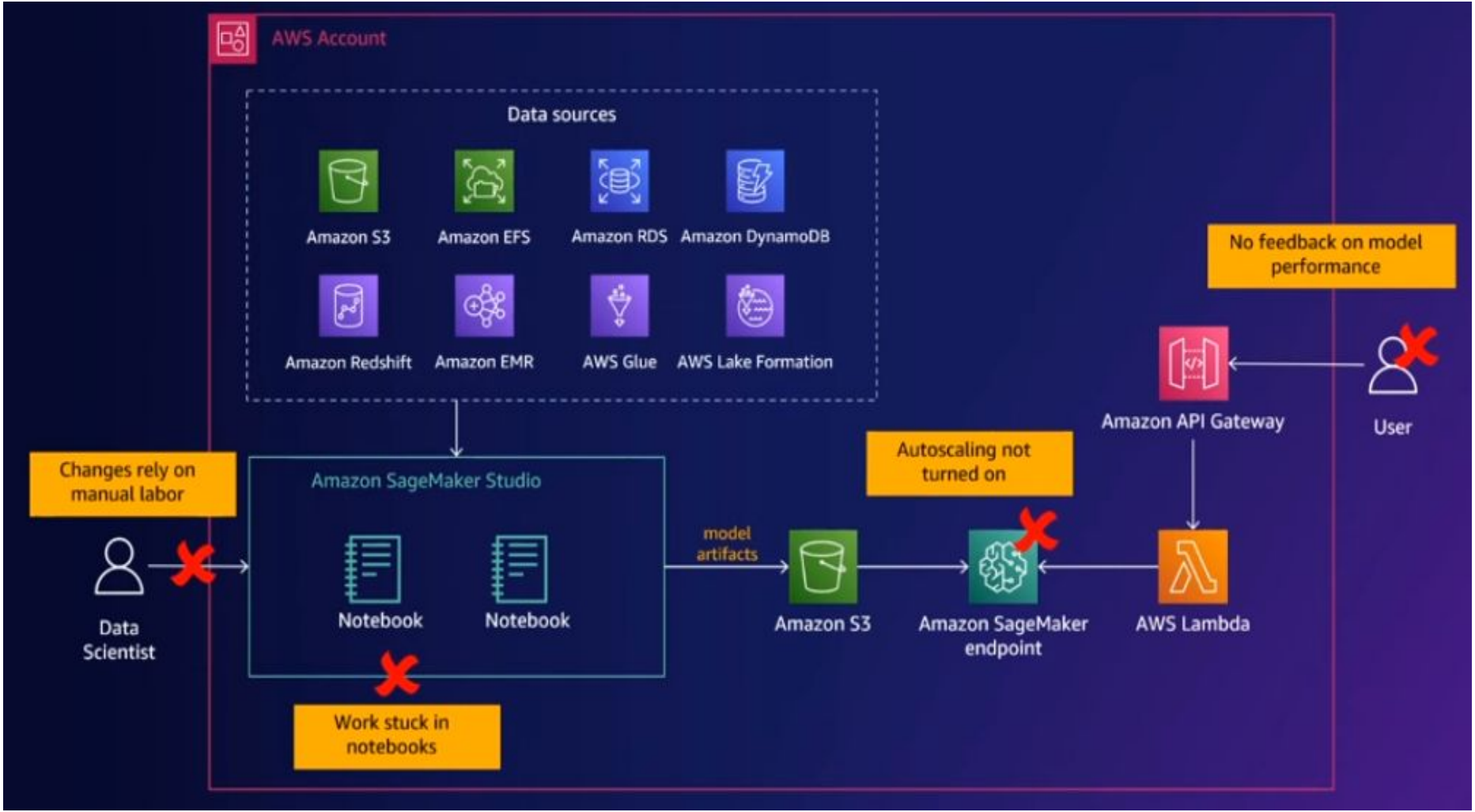
	SERVICE TYPE	DESCRIPTION	aws	Azure	Google Cloud
DATABASE	Relational DB management	Standard SQL DB (PostgreSQL, MySQL, SQL Server, etc.)	Relational Database Service (RDS), Aurora	SQL, SQL Database	Cloud SQL, Cloud Spanner
	NoSQL: Key-value	Redis-like DBs for semi-structured data	DynamoDB	Cosmos DB, Table storage	Cloud BigTable, Firestore
	NoSQL: Document	MongoDB/CouchDB-like DBs for hierarchical JSON data	DocumentDB	Cosmos DB	Firestore, Firebase Realtime Database
	NoSQL: Column store	Cassandra/HBase-like DBs for structured hierarchical data	Keyspaces	Cosmos DB	Cloud BigTable
	NoSQL: Graph	Neo4j-like DBs for connected data	Neptune	N/A	N/A
	Caching	Redis/Memcached-like memory for calculation	ElastiCache	Cache for Redis, HPC Cache	Memorystore
	Time Series DB	DB tuned for time series data	Timestream	Time Series Insights	Cloud BigTable
	Blockchain	Dogecoin, etc.	Managed Blockchain	Blockchain Service, Blockchain Workbench, Confidential Ledger	N/A

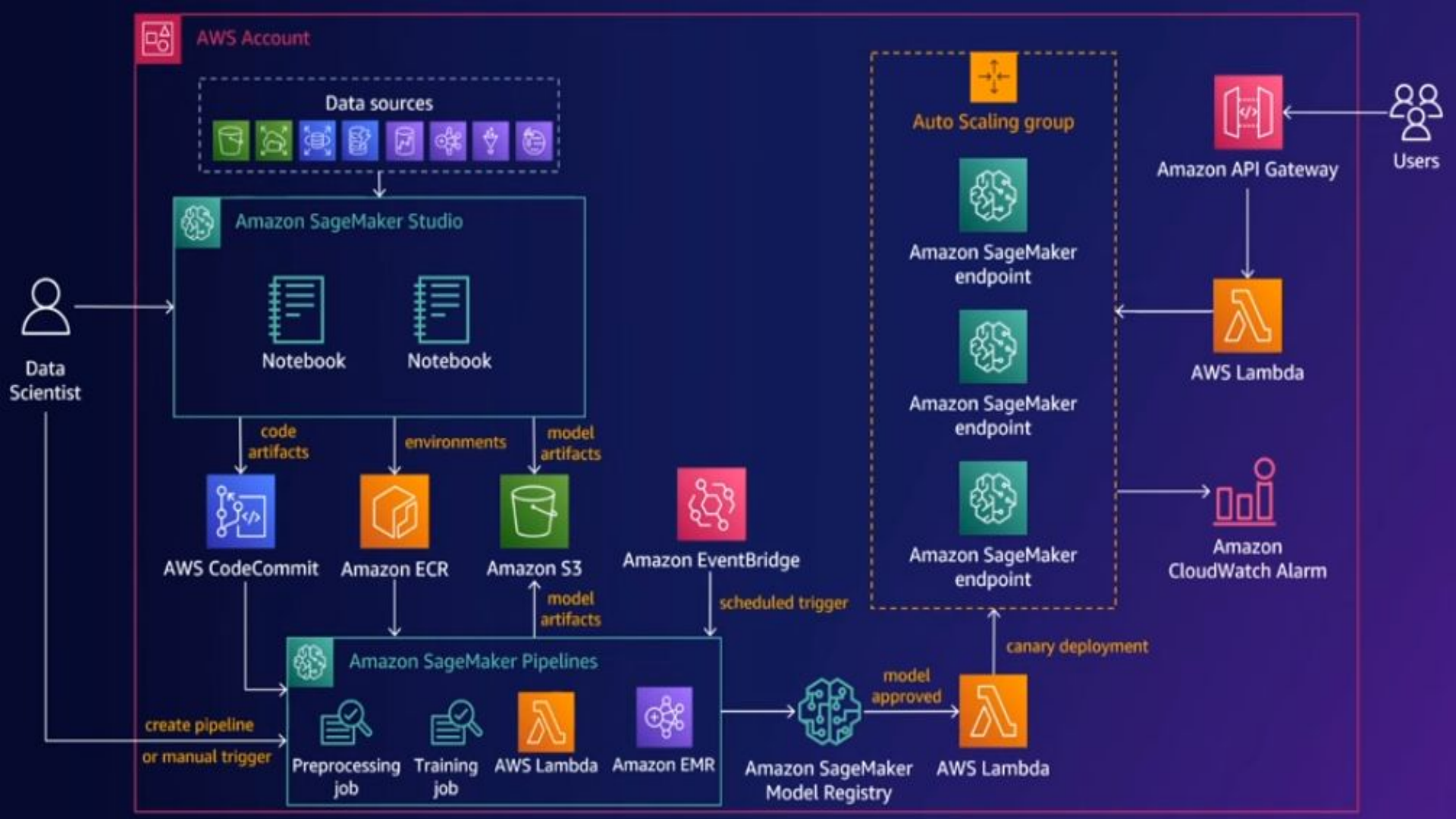
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ANALYTICS	Data Warehouse	Data Warehouse Centralized platform for all your data	RedShift	Synapse Analytics	BigQuery
	Big data platform	Run Spark, Hadoop, Hive, Presto, etc.	EMR	Data Explorer, HDInsight	Dataproc
	Business analytics	Dashboards and visualization	Quicksight, FinSpace	Power BI Embedded, Graph Data Connect	Looker, Looker Studio, Vertex AI Workbench
	Real-time analytics	Streaming data analytics	Kinesis Data Analytics, Kinesis Data Streams, Managed Streaming for Kafka	Stream Analytics, Event Hubs	Dataflow, Pub/Sub, Datastream
	Extract-Transform-Load (ETL)	Preprocessing and importing data	Glue, Kinesis Data Firehose, SageMaker Data Wrangler	Data Factory	Data Fusion, Dataflow, Dataproc, Dataprep by Trifacta
	Workflow orchestration	Build data and model pipelines	Data Pipeline, Managed Workflows for Airflow	Data Factory	Cloud Composer
	Data lake creation	Import data into a lake	Lake Formation	Data Share	Cloud Storage
	Managed search	Enterprise search	CloudSearch, OpenSearch Service, Kendra	Cognitive Search	Cloud Search
	Data Catalog	Metadata management	Glue Data Catalog	Purview, Data Explorer	Data Catalog

	SERVICE TYPE	DESCRIPTION	aws	Azure	Google Cloud
COMPUTE	**Virtual machines**	Software-emulated computers	Elastic Compute Cloud (EC2)	Virtual Machines	Compute Engine
	Spot virtual machines	Cost-effective VMs	EC2 Spot Instances	Spot Virtual Machines	Spot VMs
	Autoscaling	Autoscaling Adjust resources to match demand	EC2 Auto Scaling	Virtual Machine Scale Sets	Instance Groups
	Functions as a service (Serverless computing)	Execute code chunks without worrying about infrastructure	Lambda	Functions	Cloud Functions
	Platform as a service	Manage applications without worrying about infrastructure	Elastic Beanstalk, Red Hat OpenShift on AWS	App Service, Cloud Services, Spring Cloud, Red Hat OpenShift	App Engine
	Batch scheduling	Run code at specified times	Run code at specified times Batch	Batch	Batch, Cloud Scheduler
	Isolated servers	VM on your own machine, for high security	Dedicated Instances	Dedicated Host	Sole-tenant Nodes, Shielded VMs
	On-premise/Ed	Cloud-services on your own hardware	Outposts, Snow	Modular	N/A
	On-premise/Edge devices	Cloud-services on your own hardware	Outposts, Snow Family	Modular Datacenter, Stack Hub, Stack HCI, Stack Edge	N/A
	Quantum computing	Determine if cat is alive or dead	Braket	Quantum	N/A

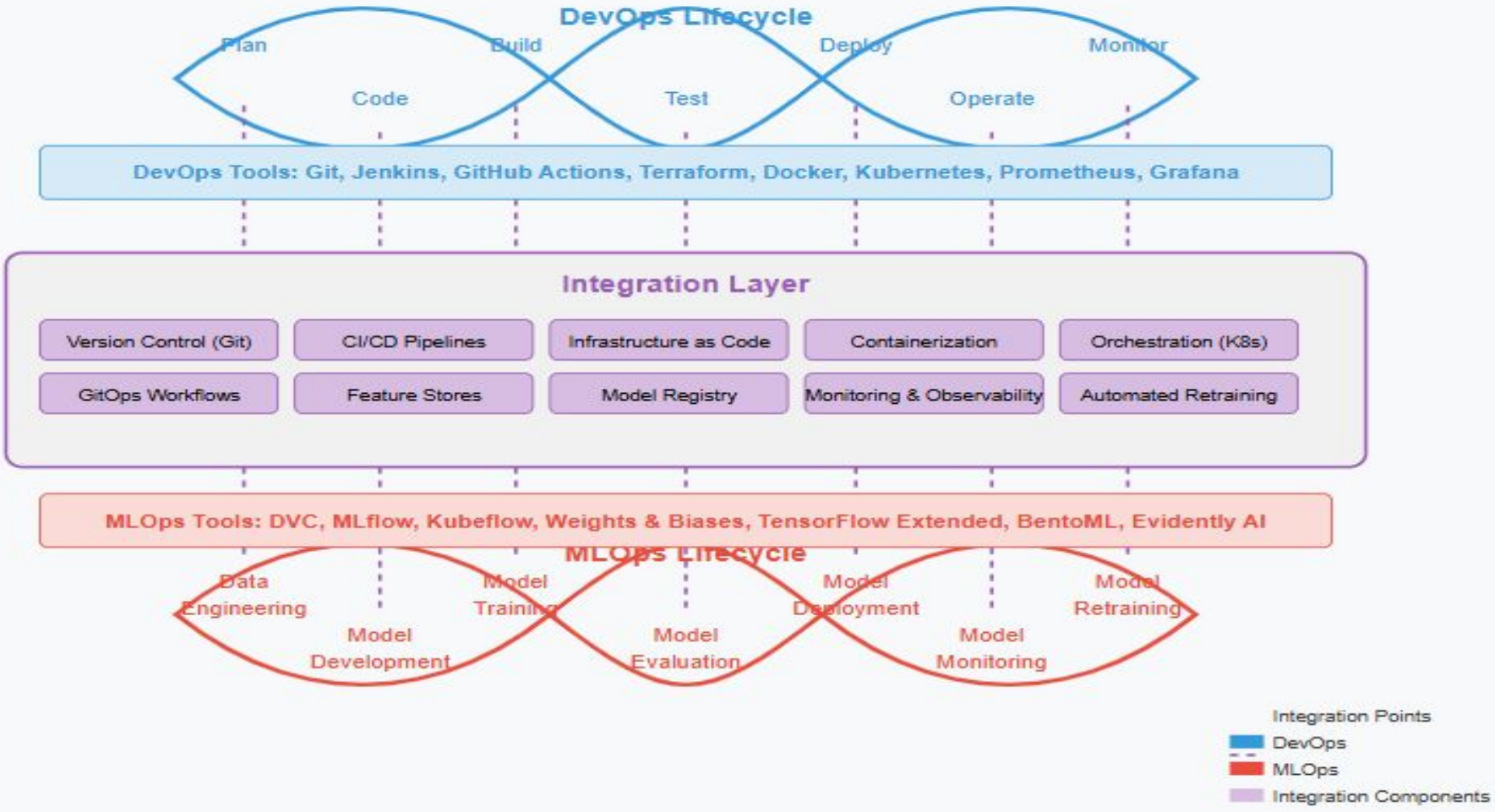
	SERVICE TYPE	DESCRIPTION	aws	Azure	Google Cloud
ML & AI	**Machine Learning**	Train, fit, validate, and deploy ML models	SageMaker	Machine Learning	Vertex AI
	Jupyter notebooks	Write data analyses and reports	SageMaker Notebooks	Notebooks	Colab
	Data science/machine learning VM	Virtual machines tailored to data work	Deep Learning AMIs	Data Science Virtual Machines	Deep Learning VM
	AutoML	Automatically build ML models	SageMaker	Machine Learning Studio, Automated ML	Vertex AI Workbench
	Natural language Processing AI	Analyze text data	Comprehend	Text Analytics	Natural Language AI
	Recommendation AI	Product recommendation engine	Personalize	Personalizer	Recommendations AI
	Document capture	Extract text from printed text & handwriting	Textract	Form Recognizer	Document AI
	Computer vision	Image classification, object detection & other AI with image data	Rekognition, Panorama, Lookout for Vision	Cognitive Services for Vision	Vision AI
	Speech to text	Speech transcription	Transcribe	Cognitive Services for Speech to Text, Cognitive Services for Speaker Recognition	Speech-to-Text
	Text to speech	Text to speech Speech generation	Polly	Cognitive Services for Text to Speech	Text-to-Speech
	Translation AI	Convert text between human languages	Translate	Cognitive Services for Speech Translation, Translator	Translation AI
	Video Intelligence	Video indexing and asset search	Rekognition Video	Video Indexer	Video Intelligence API


AWS Account
Data sources
Amazon S3
Amazon EFS
Amazon RDS
Amazon DynamoDB
Amazon Redshift
Amazon EMR
AWS Glue
AWS Lake Formation
Amazon API Gateway
User
Amazon SageMaker Studio
Notebook
Notebook
model artifacts
Amazon S3
Amazon SageMaker endpoint
AWS Lambda
Data Scientist


AWS Account
Data sources
Amazon S3
Amazon EFS
Amazon RDS
Amazon DynamoDB
Amazon Redshift
Amazon EMR
AWS Glue
AWS Lake Formation
No feedback on model performance
Amazon API Gateway
User
Changes rely on manual labor
Amazon SageMaker Studio
Autoscaling not turned on
model artifacts
Data Scientist
Notebook
Notebook
Amazon S3
Amazon SageMaker endpoint
AWS Lambda
Work stuck in notebooks


AWS Account
Data sources
Amazon SageMaker Studio
Notebook
Notebook
Data Scientist
code artifacts
environments
model artifacts
AWS CodeCommit
Amazon ECR
Amazon S3
Amazon EventBridge
scheduled trigger
model artifacts
Amazon SageMaker Pipelines
create pipeline
or manual trigger
Preprocessing job
Training job
AWS Lambda
Amazon EMR
Amazon SageMaker Model Registry
model approved
AWS Lambda
canary deployment
Auto Scaling group
Amazon SageMaker endpoint
Amazon SageMaker endpoint
Amazon SageMaker endpoint
Amazon API Gateway
Users
AWS Lambda
Amazon CloudWatch Alarm

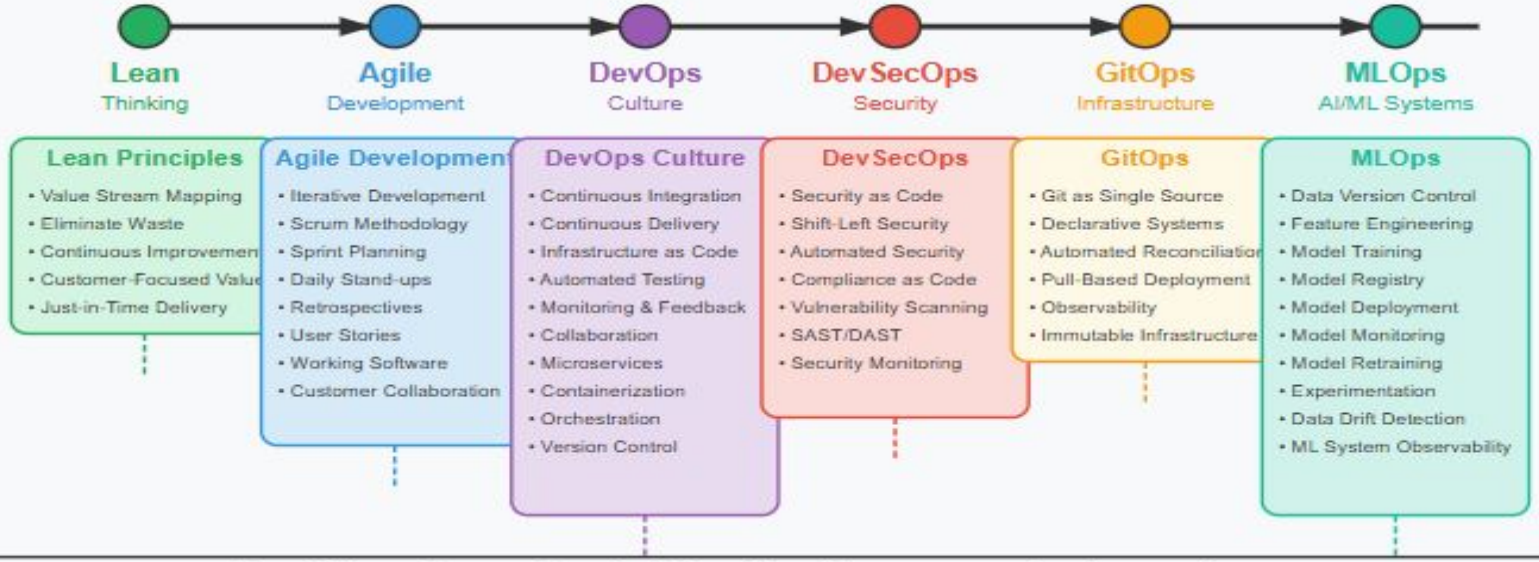
MLOps and DevOps Integration Diagram



MLOps-Specific Requirements Mapping to DevOps

DevOps Practice	MLOps Extension	Tools & Technologies
Version Control	Version Control for Code, Data, and Models	Git, DVC, Git LFS, ModelDB
CI/CD	ML Pipeline Automation Data/Model Validation	Jenkins, GitHub Actions, TFX, Kubeflow
Testing	Model Testing, Performance Validation	pytest, Great Expectations, TensorFlow Testing
Deployment	Model Serving, A/B Testing, Shadow Deployment	TF Serving, KServe, BentoML, Seldon Core
Monitoring	Model Performance, Data Drift Detection	Prometheus, Grafana, Evidently, Weights & Biases
Infrastructure as Code	ML-Specific Infrastructure, GPU/TPU Resources	Terraform, Ansible, Kubernetes, SageMaker
GitOps	Declarative ML Pipelines, Model Deployment as Code	ArgoCD, Flux, Airflow, Metaflow

The Evolution from Lean Thinking to MLOps



Core Values: Automation, Collaboration, Measurement, Continuous Improvement

Evolutionary Progress: From Lean to MLOps

Value-Driven Evolution

- Focus on delivering customer value
- Elimination of waste at each stage
- Shorter feedback loops
- Increased efficiency in delivery
- Adaptation to changing requirements
- Continuous improvement mindset

Technical Progression

- Increasing automation at each phase
- Evolution from manual to automated processes
- More sophisticated tooling & practices
- Integration of specialized concerns
- Shifting from reactive to proactive
- Building on previous foundations

Cultural Transformation

- Breaking down silos between teams
- Shared responsibility models
- Cross-functional collaboration
- Shifting mindsets & perspectives
- Balancing innovation with stability
- Everyone owns quality & delivery

The most AI Trend in 2024

1. Reality Check
2. Multi Model AI
3. Smaller Models
4. GPU Vs Cloud Cost
5. Model Optimization
6. Custom Local Models
7. Virtual Agents
8. Regulator
9. Shadow AI

Reality Check

1. Realistic Expectations: The year 2024 signifies a shift towards setting more realistic expectations for AI technologies.
2. Gen AI Adoption: Gen AI is gaining widespread attention, attracting users across various sectors to leverage AI capabilities.
3. Understanding AI: People will deepen their understanding of AI's capabilities and limitations, leading to more informed decisions about its integration into workflows.
4. Refined Understanding: A more nuanced understanding of AI-powered solutions will emerge, enabling users to discern the true potential of these technologies.
5. Complementary Nature: It will become evident that AI is not meant to replace existing tools but rather complement them in enhancing productivity and efficiency.
6. Integration into Workflows: AI will be integrated seamlessly into everyday workflows, showcasing its capabilities while serving as a supportive tool in achieving tasks and goals.

Multi Model AI

1. Versatile Models: Multi-Model AI handles inputs from various sources like text, images, and audio.
2. Integrated Information: Integrates information from different modalities to understand and generate outputs.
3. Enhanced Interactivity: Users can interact using different input formats for more meaningful responses.
4. Bidirectional Processing: Converts between modalities, like text-to-speech or image descriptions.
5. Wide Applications: Useful across domains for tasks like content generation, translation, and recommendation.
6. Research Advances: Recent models like Gemini and ChatGPT 4 push boundaries in understanding and generating content across modalities.

Smaller Models

1. Energy Consumption Concerns: Massive models like ChatGPT 3 consume significant energy during both training and inference, posing environmental and cost challenges.
2. Resource Intensiveness: While smaller models are less resource-intensive, they still offer competitive performance. Innovations focus on maximizing output with fewer parameters.
3. Parameter Reduction: Recent models like GPT-4 aim to balance performance and efficiency, with rumored parameter sizes of 1.76 trillion. However, open-source models achieve success with 3 to 17 billion parameters.
4. MixTrial Release: MixTrial, with 8 neural networks each having 7 billion parameters, outperforms larger models like Llama 2 while being more cost-effective and faster in inference. It's suitable for running locally on various devices, offering practicality and accessibility.

GPU and Cloud Cost

1. Driver of Smaller Models: The trend towards smaller models is primarily driven by necessity and entrepreneurial vigor. Larger models incur higher GPU costs for both training and inference.
2. Cloud Infrastructure Pressure: Many AI innovators and adopters rely on cloud infrastructure rather than maintaining their own. This reliance puts upward pressure on cloud costs as providers update and optimize their infrastructure to support Generative AI demands.
3. Optimization Benefits: If AI models were more optimized, they would require less compute power, leading to reduced GPU and cloud costs. This optimization is crucial for sustainability and cost-effectiveness in AI development and deployment.

Model Optimization

1. Adoption of Techniques: There's been increased adoption of techniques for optimizing pre-trained models. These techniques include quantization and Low Rank adaptation (LoRA).
2. Quantization: Quantization involves reducing the precision used to represent model data points, such as lowering from 16-bit floating data points to 8-bit integers. This reduces memory usage and speeds up inference.
3. Low Rank Adaptation (LoRA): LoRA entails freezing pre-trained model weights and adding trainable layers in each transformer block. This technique reduces the number of parameters needing updates, leading to more efficient fine-tuning and reduced memory requirements.
4. Emerging Techniques: Expect to see more model optimization techniques emerging this year to further enhance the efficiency and effectiveness of AI models.

Custom Local Models

1. Customization Opportunity: Open-source models provide the opportunity to develop powerful custom AI models tailored to an organization's proprietary data and specific needs.
2. Local Inference and Training: Keeping AI inference and training local helps avoid the risk of proprietary data and sensitive personal information being used to train closed-source models or being accessed by third parties.
3. RAG and Similar Techniques: Techniques like RAG (Retrieval-Augmented Generation) allow accessing relevant information without storing it directly within the LLM model, reducing model size and enhancing efficiency.

Virtual Agents

1. Task Automation Focus: Virtual agents go beyond traditional chatbots for customer experiences. They're focused on task automation, handling tasks like making reservations, completing checklists, or connecting to other services.
2. Versatile Functionality: Virtual agents have diverse capabilities and are designed to get things done for users, expanding beyond simple conversational interactions.

Regulation

1. European AI Regulation: In December, the European Union agreed on the Artificial Intelligence Act, marking a significant step towards regulating AI technologies.
2. Focus on AI Training: The regulation addresses the training of AI models, particularly those used for material and content generation. This remains a contentious issue, and further developments are expected in this area.

Shadow AI

1. Shadow AI involves employees using AI tools at work without official approval or oversight from the IT department.
2. Studies suggest that a significant portion, up to 90%, of employees engage in Shadow AI activities due to the absence of corporate AI policies.
3. The lack of established AI policies can lead to various issues, including concerns related to security, privacy, and compliance.
4. Employees may inadvertently expose sensitive data, such as sales figures or trade secrets, to public AI models, posing a risk to data security.
5. Another risk is the use of copyrighted material to train proprietary AI models, potentially resulting in legal consequences for the company.
6. The dangers of Shadow AI highlight the need for organizations to develop clear AI policies and guidelines.
7. It underscores the importance of educating employees about the responsible use of AI technologies and the potential risks associated with Shadow AI.
8. By implementing robust AI policies and ensuring employee compliance, organizations can mitigate the risks posed by Shadow AI and promote responsible AI usage in the workplace.

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